

Minseo Kim

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RESEARCH INTERESTS

AI & Machine Learning Systems, Efficiency, HW-SW Co-Design

EDUCATION

- **Seoul National University** Mar 2020 – Aug 2026 (Expected)
B.S. in Computer Science and Engineering, Summa cum laude (expected) Seoul, South Korea
 - GPA: **4.20 / 4.30 (major)**, 4.11 / 4.30 (overall)
 - 2-year absence to fulfill military duty (Oct. 2021 - Jul. 2023)
- **University of California, Berkeley** Jun 2025 – Aug 2025
Visiting student Berkeley, CA, USA
 - GPA: 4.00 / 4.00; Courses: Computer Security (A+), Introduction to Artificial Intelligence (A+)
- **Korea Minjok Leadership Academy** Mar 2017 – Feb 2020
Secondary Education, First class honor in STEM field (1st / 73) Gangwon-do, South Korea

PUBLICATIONS

C=CONFERENCE, S=IN SUBMISSION

* Equal contribution

- [C.1] Taebaek Hwang*, Minseo Kim*, Gisang Lee, Seonuk Kim, Hyunjun Eun (2025). **KRETA: A Benchmark for Korean Reading and Reasoning in Text-Rich VQA Attuned to Diverse Visual Contexts**. In Proceedings of EMNLP 2025. [[Paper](#)] [[Project Page](#)][[Hugging Face](#)]
- [C.2] Bonggeun Sim, Yushin Kim, Minseo Kim, Yeonhong Park, Jae W. Lee (2025). **InstANNS: Scalable Approximate Nearest Neighbor Search via Cost-Efficient In-Storage Processing**. In Proceedings of CIKM 2025. [[Paper](#)]
- [S.1] Minseo Kim, Chenfeng Xu, Coleman Hooper, Harman Singh, Ben Athiwaratkun, Ce Zhang, Kurt Keutzer, Amir Gholami (2025). **CDLM: Consistency Diffusion Language Models for Faster Sampling**. Submitted to MLSys 2026. [[arXiv](#)]

PREPRINTS

- [1] Minseo Kim, Coleman Hooper, Aditya Tomar, Chenfeng Xu, Mehrdad Farajtabar, Michael W. Mahoney, Kurt Keutzer, Amir Gholami (2025). **Beyond Next-Token Prediction: A Performance Characterization of Diffusion versus Autoregressive Language Models**. *arXiv preprint*. [[arXiv](#)]

RESEARCH EXPERIENCE

- **Berkeley Artificial Intelligence Research (BAIR) Lab**  Jun 2025 – Jan 2026
Undergraduate Visiting Researcher (Advisor: Prof. Kurt Keutzer) Berkeley, CA, USA
 - LLM inference on compute-in-memory (CIM) hardware
 - Proposed and implemented **KV-cache compression** via offline dictionary learning and online sparse coding for memory-constrained CIM deployments.
 - Integrated adaptive hierarchical sparsity and query-aware KV reconstruction, reducing dynamic memory loads by $15\times$ while maintaining performance across LongBench tasks.
 - Efficient inference in **Diffusion Language Models (DLMs)**
 - Characterized DLMs via GPU roofline analysis to evaluate implications for block-wise and batched decoding, identifying excessive refinement steps and caching incompatibility as primary bottlenecks.
 - Developed a consistency-distilled DLM with block-causal attention to enable KV caching, reducing refinement steps by $3.4\times\text{--}7.9\times$ and latency by $3.6\times\text{--}14.5\times$ while maintaining accuracy on math/code benchmarks.
- **SNU Architecture and Code Optimization Lab**  Jan 2025 – Feb 2025
Undergraduate Researcher (Advisor: Prof. Jae W. Lee) Seoul, South Korea
 - Integrated SPDK into the existing ANNS system and optimized it, resulting in a $2.15\times$ increase in throughput.
 - Built the host $\leftrightarrow$ SSD interface for a new NVMe command enabling in-storage PQ filtering, cutting host I/O.

WORK EXPERIENCE

- **FuriosaAI** Jan 2026 – Jul 2026 (expected)
AI Algorithm Research Intern Seoul, South Korea
 - TBD
- **Deeping Source Inc.**  Jul 2024 – Aug 2024
ML Engineering Intern Seoul, South Korea
 - Added a gaze-vector inference capability to an existing ML model for pedestrian attribute recognition, achieving a 90% soft-hit rate.
 - Generated training data using IMU sensors and built a robust data pipeline for large-scale data acquisition.

PROJECTS

- **Open-Source Contribution, Samsung NNTrainer** Oct 2025 – Nov 2025
Tools: C++
 - Enhanced NNTrainer’s on-device training stack by adding gradient checkpointing and lightweight optimizers (Lion, Sophia) for improved memory and training efficiency.
- **ARC-AGI (Abstraction & Reasoning Corpus) Solver** Mar 2025 – Jun 2025
Tools: PyTorch 
 - Intro to Deep Learning term project: built an ARC-AGI solver; placed **1st** among 35 teams on the hidden evaluation.
 - Fine-tuned Qwen3-4B with LoRA, curated data augmentation, and a custom lm_head; two test-time scaling methods (test-time training and grid-wise voting). (Technical report: [link](#))
- **KRETA: Korean Text-Rich VQA Benchmark & VLM Fine-Tuning** Oct 2024 – Feb 2025
Tools: PyTorch  
 - Collected Korean text-rich image datasets and fine-tuned LLaVA-Onevision to strengthen Korean capability.
 - Built an end-to-end generation pipeline and released a high-quality Korean text-rich VQA benchmark.
- **Art College Graduation Exhibition Sales Website** Sep 2024 – Dec 2024
Tools: Django, React, Supabase, AWS  
 - Designed a Supabase-based database, implemented RESTful APIs with Django, integrated a KakaoPay module for payment processing, and deployed the service on AWS.
 - Collaborated with the Metal Craft and Ceramics departments at SNU for product registration.
- **Window Software Vulnerability Research & Exploitation** Sep 2023 – Feb 2024
Tools: Windbg, IDA, Pwndbg, Fuzzer, x86-64 Assembly
 - Completed intensive training at WhiteHat School (web, system, cloud security, forensics) and researched Windows application vulnerabilities using static/dynamic analysis and fuzzing.
 - Responsibly disclosed **three CVE-registered systems vulnerabilities** via ZDI ([CVE-2024-11510](#), [CVE-2024-11511](#), [CVE-2024-11512](#)); coordinated with vendors to ensure patch releases.

HONORS AND AWARDS

- **National IT Industry Promotion Agency (NIPA) President’s Award - 2025 AI Chip Contest** Dec 2025
 - Grand Prize (top award) in the AI semiconductor application track for optimizing vision pre-processing workloads on Furiosa/Rebellions NPUs (KRW 10M prize).
- **Gaheonsindo Foundation Scholarship (Full Undergraduate Tuition)** Mar 2024 – Present
- **SNU Specialized Semiconductor College Scholarship** Feb 2024 – Present
- **Accelerator Programming Winter School Outstanding Graduate** Feb 2025
 - Secured 1st place in CUDA-based heterogeneous and accelerator computing competitions.
- **Ministry of Science and ICT Award - Whitehat School Outstanding Graduate** Mar 2024
 - Ranked 1st out of 309 participants (KRW 3M prize).
- **Merit-based Scholarship** 2021, 2023

LEADERSHIP & TEACHING

- **President, Guardian (SNU CSE Security Research Club)** Aug 2024 – Aug 2025
 - Organized and delivered weekly seminars for new members on system hacking, reversing, web security, and cryptography; designed onboarding curriculum and mentorship.
- **Presenter (28th Winter Hacking Camp) – Accepted Talk** Feb 2024
 - Delivered a 1-hour instructional session on Windows software vulnerability detection and analysis to 60 participants.
- **Team WWW Lead (Whitehat School)** Oct 2023 – Jan 2024
 - Led an outstanding project team; discovered three CVE-assigned vulnerabilities and concluded with a poster presentation.

SKILLS

- **Programming Languages:** Python, x86-64 Assembly, C, C++, Java
- **AI/ML:** PyTorch, CUDA, Triton, OpenCV, ClearML
- **DevOps & Version Control:** Git, Docker
- **Other Tools & Technologies:** IDA, WinDbg, Pwndbg, Fuzzer, Unity, React, Django, Supabase, MySQL

LANGUAGE PROFICIENCY

Korean (Native), English (Fluent)